

TRAPPIST-1 (35 points)

Aliens have found out that Earth's astronomers discovered planets in the TRAPPIST-1 system by observing numerous transits. They have used their flying saucer (similar to the one you were in for the observation round) to take you to the 5th planet (designated *f*) of TRAPPIST-1, and have asked you to show them the methods Earthlings use to uncover the parameters of the system. A clock displaying time in Earth hours will be visible. The whole presentation lasts 520 h (1 s represents 1 h).

Based on your observations (you can use the space on the last sheet for observing notes),

(a) determine the following quantities for the planet you are on (use Earth hours for the times):

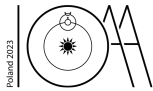
3.a (7 pt)

i.	length of the sidereal day [h]	
ii.	orbital period [h]	
iii.	length of the 'solar' day [h]	
iv.	circular orbit	YES / NO
v.	obliquity (axial tilt) [°]	

(b) and the following quantities for each planet *b*, *c*, *d* and *e*:

3.b (16 pt)

	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
synodic period [h]				
maximum elongation [°]				



(c) calculate the orbital period in hours and the semi-major axis in τ (where 1 τ = "TRAPPIST-1 f astronomical unit" = the semi-major axis of the orbit of TRAPPIST-1 f) of each planet:

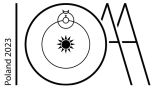
3.c (8 pt)

	b	c	d	e
orbital period [h]				
semi-major axis [τ]				

(d) The term 'gravitational resonance' is used to describe the phenomenon when ratio of the orbital periods of two planets in a system is close to the ratio of two integers. The table below lists some of the resonances observed in the TRAPPIST-1 system. Find which pair(s) of planets correspond to each of the listed resonances if any.

3.d (4 pt)

Resonance	Pair of planets
3:2	
8:5	
5:3	
8:3	
4:1	
6:1	



Observing notes

Star (TRAPPIST-1*a*)

TRAPPIST-1*b*

TRAPPIST-1*c*

TRAPPIST-1*d*

TRAPPIST-1*e*